MitoShape: An Image Analysis Tool for High-Content Quantification of Mitochondrial Morphology Across Cell Types and Physiological Conditions

MitoShape is a Fiji/ImageJ-based analysis pipeline developed by Juan Diego Hernández Camacho under the supervision of the Timothy Wai.

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The primary objective of these macros is to **quantify mitochondrial shape descriptors in cultured cells using fluorescence microscopy images**. For the development and validation of this tool, images were acquired using the **Operetta CLS High-Content Analysis System (PerkinElmer)** with a **63× objective**. The first step of the pipeline consists of **automatically organizing raw images according to their acquisition channel**. This is achieved using a Fiji/ImageJ macro that classifies images into separate folders based on channel identifiers embedded in the file names (e.g., *ch1*, *ch2*, *ch3*), corresponding to the different fluorescence channels.

For the development of this tool, we analyzed **SARS-CoV-2–infected cells**, which resulted in **heterogeneous cell populations within the same well**. These populations were distinguished by immunofluorescence based on the expression of the **SARS-CoV-2 nucleocapsid (N) protein**, allowing discrimination between **infected (N-positive)** and **non-infected (N-negative)** cells. To separate these populations, the nucleocapsid signal was used as a **differential marker**.

To achieve this, a **multi-step Fiji/ImageJ macro** was implemented for segmentation and mask processing, tailored to distinguish nucleocapsid (NM) and mitochondrial signals. The macro performs the following operations:

- **Segmentation of the SARS-CoV-2 nucleocapsid channel** using a fixed intensity threshold, followed by morphological cleanup and noise removal to generate a refined binary mask.
- **Segmentation of the mitochondrial channel** using a Difference of Gaussians (DoG) filtering approach to enhance structures of interest, followed by automatic thresholding.
- **Generation of backup copies** of the processed masks to preserve intermediate results.
- **Inversion of the nucleocapsid masks**, enabling downstream analyses such as exclusion-based quantification or spatial filtering.

The third step of the macro performs a **pairwise subtraction followed by particle analysis** between the SARS-CoV-2 nucleocapsid channel (ch2) and the mitochondrial channel (ch3), in order to quantify the remaining mitochondrial structures after exclusion of nucleocapsid signal. The workflow begins by loading the corresponding image folders, applying filtering to retain only TIFF files, and sorting them to ensure correct pairing. Each nucleocapsid image is matched to its corresponding mitochondrial image based on a shared filename prefix. For each image pair, the Fiji **Image Calculator** is used to perform a pixel-wise subtraction (**mitochondria − nucleocapsid**). This operation removes overlapping signal and isolates

mitochondrial regions that are spatially independent of nucleocapsid-positive areas, generating a refined image for downstream quantitative analysis.

The resulting images are then converted into binary masks, on which particle analysis is performed to detect and quantify individual structures above a defined size threshold. The macro systematically saves all outputs into a dedicated folder, including the subtracted images, segmentation outlines, per-particle measurements, and summary statistics. This structured export enables downstream quantitative analysis of mitochondrial features that are spatially independent of the SARS-CoV-2 nucleocapsid signal.

The final macro is a **post-processing utility that consolidates multiple CSV result files into a single dataset for downstream statistical analysis**.